A Prefactor Coefficient Law for the Gauss–Legendre AGM Algorithm

A refined diagnostic asymptotic

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Abstract

For the lower Gauss–Legendre approximants π_n, consider the exact, reference-dependent diagnostic weight $\alpha_n = -(\pi_{n+1} - \pi)/(\pi_{n+1} - \pi_n)$. Using the classical two-term theta-function error expansion recorded by Brent, we recover $\alpha_{n+1}/\alpha_n^2 \to 1/2$ as a direct corollary and sharpen the limit by an explicit rational prefactor model. With $x_n = 2^{n+1}\pi$, we obtain

$$\frac{\alpha_{n+1}}{\alpha_n^2} = \frac{(4x_n - 1)(x_n - 1)^2}{(2x_n - 1)^3} + O(\mathrm{poly}(2^n)e^{-x_n}) = \frac{1}{2} - \frac{3}{8x_n} + O(x_n^{-2}).$$

This explains the observed approach to 1/2 from below. The weight depends on the unknown target π and is not an implementable acceleration method. We retain a separate numerical observation concerning numerator-channel weights, without claiming its proof.

1 AGM indexing and the diagnostic coefficient

Start with $a_0 = 1$, $b_0 = 1/\sqrt{2}$, $t_0 = 1/4$, and $p_0 = 1$, and use

$$\begin{aligned} a_{n+1} &= (a_n + b_n)/2, & b_{n+1} &= \sqrt{a_n b_n}, \\ t_{n+1} &= t_n - p_n(a_n - a_{n+1})^2, & p_{n+1} &= 2p_n. \end{aligned}$$

Throughout this paper the lower approximant is

$$\pi_n = \frac{(a_n + b_n)^2}{4t_n} = \frac{a_{n+1}^2}{t_n}, \qquad \varepsilon_n = \pi_n - \pi < 0.$$

The index on π_n is the index of a_n, b_n, t_n appearing on the left expression, not of a_{n+1}. Define

$$\alpha_n = -\frac{\pi_{n+1} - \pi}{\pi_{n+1} - \pi_n} = -\frac{\varepsilon_{n+1}}{\varepsilon_{n+1} - \varepsilon_n}.$$

Thus $\pi = \pi_{n+1} + \alpha_n(\pi_{n+1} - \pi_n)$ holds *by definition*. Since evaluation of α_n requires π, this coefficient is a retrospective diagnostic, not a computable error-correction rule or algorithmic speed claim.

2 Classical prefactor input

We use the following detailed expansion from Brent's theta-function analysis of the lower AGM approximant [1]:

$$\pi - \pi_n = (2^{n+4}\pi^2 - 8\pi)e^{-\pi 2^{n+1}} + O(2^n e^{-\pi 2^{n+2}}). \tag{1}$$

The O notation in (1) denotes a signed remainder with bounded magnitude; no sign is asserted for that remainder. This detailed two-term expansion, rather than the original 1976 complexity analysis [2], is the asymptotic input used below. Put $x_n = 2^{n+1}\pi$ and $B_n = -8\pi(x_n - 1)$, so that

$$\varepsilon_n = B_n e^{-x_n} + O(2^n e^{-2x_n}), \qquad x_{n+1} = 2x_n. \tag{2}$$

Writing $\varepsilon_n = A_n e^{-x_n}$ gives

$$A_n = B_n + O(2^n e^{-x_n}), \qquad A_{n+1}/A_n \longrightarrow 2.$$

The limit comes from Brent's expansion; it is not an independent new AGM theorem.

3 Coefficient law and sharpened correction

Lemma 1. *If $r_n = \varepsilon_{n+1}/\varepsilon_n \to 0$, then*

$$\alpha_n = \frac{r_n}{1 - r_n}, \qquad \frac{\alpha_{n+1}}{\alpha_n^2} = \frac{\varepsilon_{n+2}\varepsilon_n^2}{\varepsilon_{n+1}^3}\frac{(1 - r_n)^2}{1 - r_{n+1}}.$$

Proof. Both identities follow by substitution and cancellation. □

Theorem 1 (Prefactor coefficient limit)**.** *For the above lower Gauss–Legendre sequence,*

$$\lim_{n\to\infty} \frac{\alpha_{n+1}}{\alpha_n^2} = \frac{1}{2}.$$

Proof. By (2), $r_n \to 0$. Lemma 1 reduces the ratio asymptotically to $\varepsilon_{n+2}\varepsilon_n^2/\varepsilon_{n+1}^3$. The exponential factors cancel exactly because $x_{n+2} + 2x_n - 3x_{n+1} = 0$. The remaining ratio tends to $2(1/2)^2 = 1/2$, since $A_{n+1}/A_n \to 2$. □

Corollary 1 (Finite-index rational prefactor law)**.** *Let $x = x_n = 2^{n+1}\pi$. Then*

$$\frac{\alpha_{n+1}}{\alpha_n^2} = \frac{(4x - 1)(x - 1)^2}{(2x - 1)^3} + O(\mathrm{poly}(2^n)e^{-x}), \tag{3}$$

and hence

$$\frac{\alpha_{n+1}}{\alpha_n^2} = \frac{1}{2} - \frac{3}{8x} + O(x^{-2}) = \frac{1}{2} - \frac{3}{2^{n+4}\pi} + O(4^{-n}). \tag{4}$$

Also the rescaled error prefactors obey

$$\frac{A_{n+1}}{A_n} = 2 + \frac{1}{x - 1} + O(\mathrm{poly}(2^n)e^{-x}).$$

Proof. Use $B_n = -8\pi(x - 1)$, $B_{n+1} = -8\pi(2x - 1)$, and $B_{n+2} = -8\pi(4x - 1)$ in Lemma 1. The pure prefactor quotient is exactly the rational expression in (3). By (2), the relative errors in the B_k are exponentially small, and $r_n = O(e^{-x}\,\mathrm{poly}(2^n))$; their effect on the diagnostic quotient is likewise exponentially small. Expansion in $1/x$ gives $1/2 - 3/(8x) + O(x^{-2})$. The prefactor ratio follows from $B_{n+1}/B_n = (2x - 1)/(x - 1)$. □

Remark 1. Equation (4) describes the eventual approach to 1/2 from below. The finite-n rational expression is an asymptotic model, not an exact identity for the numerical AGM coefficient. Exponentially small corrections remain.

4 Numerical verification

The following figures are retained from the original manuscript; they were reported with 500-decimal-digit `mpmath` calculations and the above indexing. The present correction is analytical and does not replace that independent numerical record.

n	α_n
0	$4.476840443051657 \times 10^{-3}$
1	$7.278747106485182 \times 10^{-6}$
2	$2.482705775153228 \times 10^{-11}$
3	$2.988090953871379 \times 10^{-22}$
4	$4.397065311352923 \times 10^{-44}$
5	$9.594611535813435 \times 10^{-88}$
6	$4.585616315217315 \times 10^{-175}$

Table 1: Reference-dependent fitted diagnostic weights.

n	α_{n+1}/α_n^2	A_{n+1}/A_n
0	0.3631728700311416	2.3866261558728965
1	0.4686104763344650	2.0871750987545307
2	0.4847784478028806	2.0414374856886031
3	0.4924649101466708	2.0202981874959604
4	0.4962511920458477	2.0100471245391349
5	0.4981302572056423	2.0049984522335209

Table 2: Observed coefficient and prefactor ratios.

5 Numerator channel and computational scope

Set $q_n = (a_n + b_n)^2$ and

$$\beta_n = -\frac{q_{n+1} - 4\pi t_{n+1}}{q_{n+1} - q_n}.$$

Numerical Observation 1 (Numerical only)**.** The original high-precision calculation suggested $\beta_{n+1}/\beta_n^2 \to 0$. No analytical proof is provided, and the present revision does not claim independent reproduction of this secondary observation.

The static Baker N-series uses fixed rational weights to remove powers of a resolution parameter. In contrast the present α_n uses an unknown limit and is *not* an acceleration prescription. Aitken's Δ^2 extrapolation is a different, implementable three-point operation; its merit relative to taking the next AGM iterate is not asserted here without a separate error-and-cost comparison.

6 Conclusion

The limit $\alpha_{n+1}/\alpha_n^2 \to 1/2$ is a transparent consequence of Brent's lower-approximant asymptotic. Retaining his subleading -8π prefactor term yields the sharper rational model (3), which quantitatively explains the finite-index trend. This is a diagnostic comparison of error structure, not a new algorithm for π or a fixed-coefficient extrapolation method.

References

[1] R. P. Brent, "The Borwein brothers, Pi and the AGM," in *From Analysis to Visualization*, Springer Proceedings in Mathematics & Statistics 313 (2020), 323–348; also arXiv:1802.07558.

[2] R. P. Brent, "Fast multiple-precision evaluation of elementary functions," *J. ACM* 23 (1976), 242–251.

[3] J. M. Borwein and P. B. Borwein, *Pi and the AGM*, Wiley, 1987.

[4] E. Salamin, "Computation of π using arithmetic-geometric mean," *Math. Comp.* 30 (1976), 565–570.

[5] A. C. Aitken, "On Bernoulli's numerical solution of algebraic equations," *Proc. Roy. Soc. Edinburgh* 46 (1926), 289–305.